Fiscal Year:	FY 2004	**Task Last Updated:**	FY 05/11/2004
PI Name:	Marshall, Jeffrey S. Ph.D.		
Project Title:	Rivulet Dynamics with Variable Gravity and Wind Shear		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Iowa City	**State:**	IA
Zip Code:	52242	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	05/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	note - end date changed from 11/30/2003		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ettema, Robert (The University of Iowa)		
Grant/Contract No.:	NAG3-2368		
Performance Goal No.:			
Performance Goal Text:			
	The project involves examination of the effect of gravity on the formation and dynamics of liquid rivulets. The project entails both experimental and theoretical aspects. The project focuses on the dynamics of shear-driven rivulet flows on a horizontal plate and the effect of gravity normal to the plate on the rivulet breakup and the subsequent droplet flow. Computations have also been performed to examine the effect of surface contamination and normal gravity on the problem of rivulet formation from an initially straight liquid film front. The task objective is to examine the effect of gravitational force oriented normal to a plate on the different stages of rivulet development, including the initial formation of rivulets, meandering and breakup of rivulets, and the motion and interaction of liquid droplets formed from broken rivulets, in the presence of a shear flow. Rivulets play an important role in a wide variety of terrestrial applications in the energy, chemical and aerospace industries. Rivulet formation on the wings of an airplane flying in		

Task Description:	rain decrease the aerodynamic efficiency and (under freezing conditions) control the pattern of ice formation on the wings. Rivulet formation enhances heat transfer rate to the liquid stream due to the large ratio of liquid surface area to cross-sectional area. This fact has been used in a variety of cooling systems, including cooling streams of solar panels, tubes of mist-cooled heat condensers, and liquid-metal blanket cooling of fusion reactors. Mass transfer rates to rivulets is important for applications such as trickle-phase chemical reactors, flow over a catalytic pellet bed, and flow within columns with structured packing. Rivulet formation in two-phase liquid-gas flow in a tube controls the transition between annular flow to droplet flow. Rivulet formation is an undesirable feature in several coating processes that limits the minimum thickness of the coating sheet and results in inefficient and nonuniform coating of the substrate. The effect and dynamics of rivulet flows in space applications is largely unknown due to the different gravitational states involved. In particular, normal gravitational force has been found both to surpress the fingering instability of a liquid front (which preceeds rivulet formation) and decreases the sensitivity of liquid front flows to surface contamination. We thus expect that rivulet flows will be considerably more common in space-based applications than in terrestrial applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Flight experiments on-board the KC 135 were performed in May 2003 to examine wind-driven rivulet break-off and droplet flow in microgravity and partial gravity conditions (0.1g, 0.16g, 0.26g, 0.38g). The experiments show that a wind-driven rivulet exhibits a periodic break-off, leading to a droplet flow state, under a wide range of air and water flow rates and gravitational conditions. However, the rivulet break-off frequency is highly sensitive to the normal gravity acting on the substrate surface. The break-off frequency generally increases with decrease in gravitational acceleration, and in fact rivulet break-off in terrestrial gravitational conditions occurs with an order of magnitude lower frequency than break-off in zero gravity with the same water and air flows rates. The ensuing liquid droplets in microgravity are relatively small and nearly hemispherical in shape, whereas in terrestrial gravity the droplets are much larger and become highly elongated in the transverse flow direction. In terrestrial gravity the droplets are observed to bifurcate as they advect downstream, leading to a condition sufficiently far downstream where droplets with many different sizes and forward velocities exist in a highly disordered state, dominated by frequent droplet collisions and bifurcations. These droplet bifurcations and collisions are not observed in microgravity, and instead the droplets move downstream as intact individuals with approximately the same size and velocity. However, an interesting "inch-worming" motion is observed for the microgravity droplets under certain condition. In this inch-worm motion, the liquid within a droplet will slosh back and forth, from the rear to the front of the droplet, and the droplet will progress downstream in a series forward lurches each time the liquid sloshes forward. This inch-worming motion is not observed in terrestrial gravity, nor in any of the partial gravity conditions examined in our experiments. Models have been developed for many of the experimentally-observed flow features, which provide either analytical predictions or allow a collapsing of the experimental data. These features include rivulet break-off frequency, critical height and surface area of the rivulet "head", rivulet head "creep" velocity, and droplet advection speed. These models have been shown to compare well to the experimental data. Additional purely computational work is on-going on the problem of formation of rivulets on "natural" surfaces, where a random variation in contact angle exists. This work examines the effect of contact angle variation on the fingering instability of a driven liquid film front. In future work, we plan to examine more realistic random patterns of contact angle variation and to study the effect of liquid inertia on the sensitivity of the film to contact angle variation.
Bibliography Type:	Description: (Last Updated: 02/27/2004)